Time Series HW6

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March 2, 2026

Task 1

Scenario 1: Large difference in AR coefficient
True break: 100
Estimated break: 100 ($F = 45.35$)
$\rightarrow$ Null hypothesis of no break **REJECTED** (break detected).

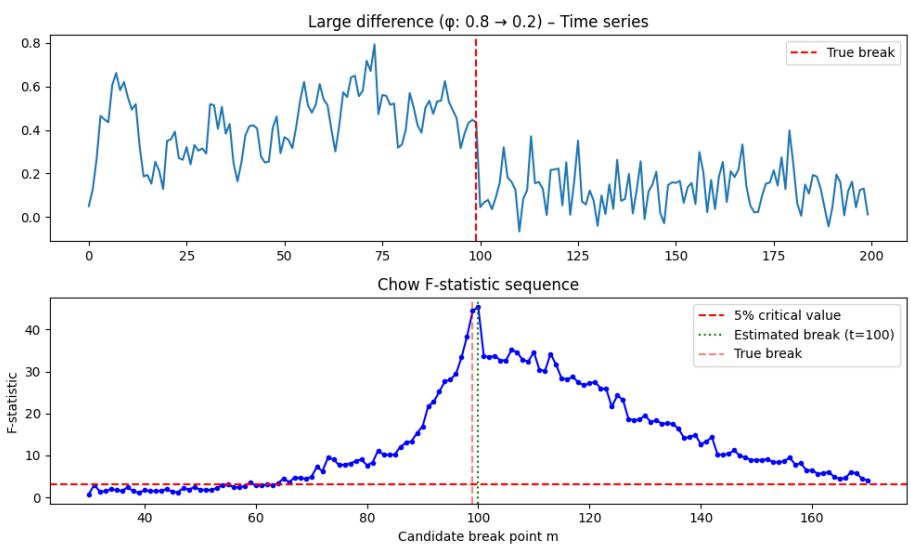

Figure 1: Large difference in AR coefficient

Scenario 2: Small difference in AR coefficient and intercept
True break: 100
Estimated break: 113 ($F = 5.53$)
$\rightarrow$ Null hypothesis of no break **REJECTED** (break detected).

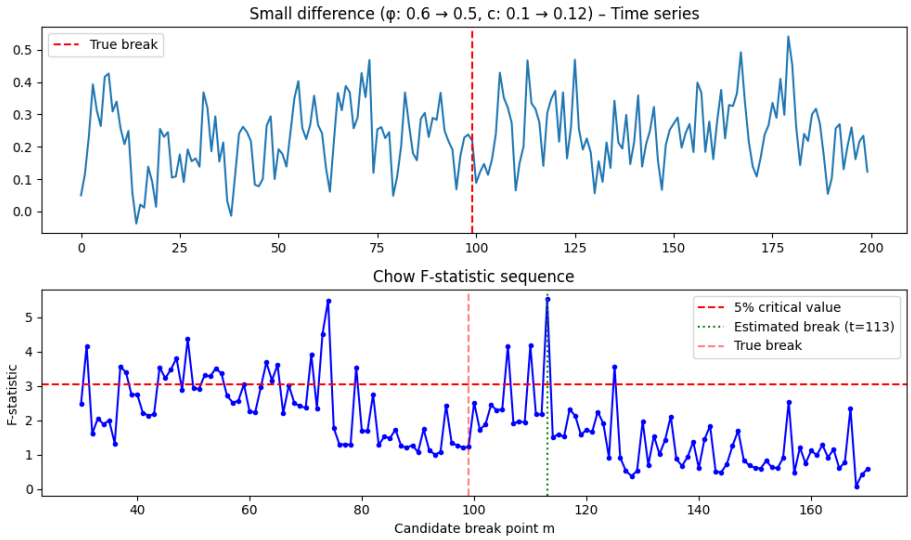

Figure 2: Small difference in AR coefficient and intercept

Scenario 3: Longer series ($T = 400$) with small difference
True break: 200
Estimated break: 209 ($F = 8.75$)
$\rightarrow$ Null hypothesis of no break **REJECTED** (break detected).

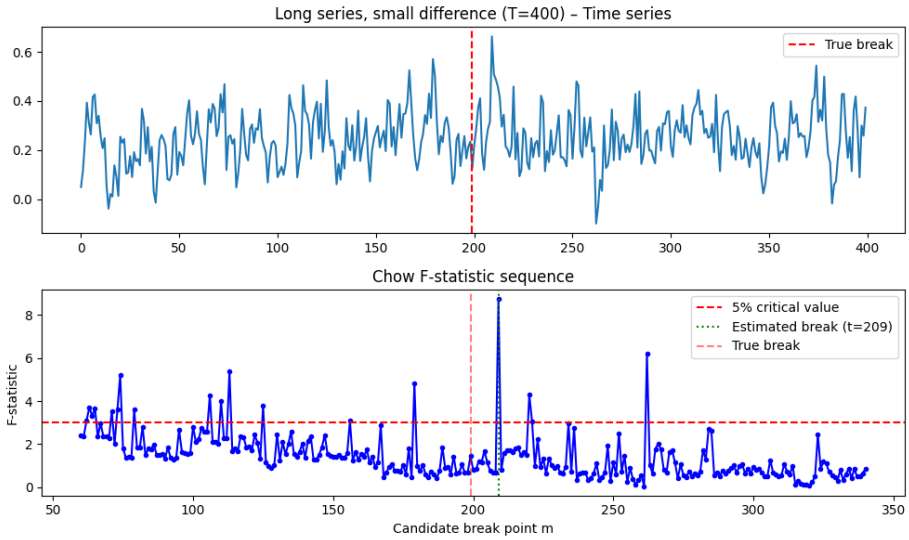

Figure 3: Longer series ($T = 400$) with small difference

Task 2

Detailed test outputs:

- **Testing AR(1)**
  Sup-F = 33.0380 (p-value: 0.0000)
  Ave-F = 6.7089
  Exp-F = 12.7966
  Most likely break index (observation): 259

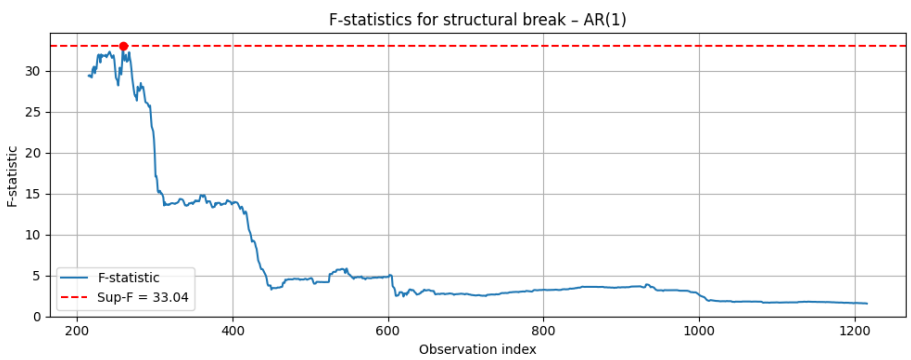

Figure 4: AR(1)

- **Testing AR(4)**
  Sup-F = 14.2366 (p-value: 0.0142)
  Ave-F = 2.3514
  Exp-F = 3.8083
  Most likely break index (observation): 241

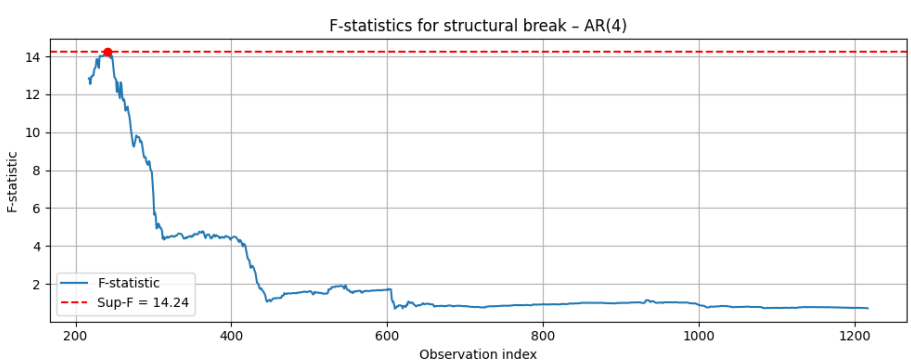

Figure 5: AR(4)

- **Testing AR(5)**

Sup-F = 12.3869 (p-value: 0.0539)
Ave-F = 2.0159
Exp-F = 2.9936
Most likely break index (observation): 241

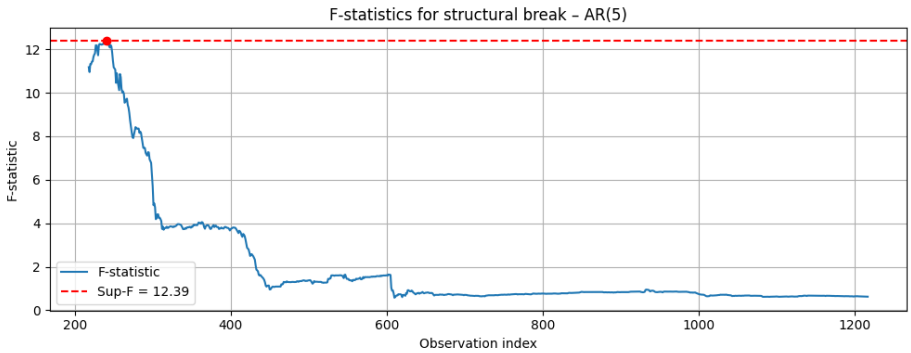

Figure 6: AR(5)

- **Testing ARMA(1,1)**
  Sup-F = 10.6055 (p-value: 0.1568)
  Ave-F = 1.7339
  Exp-F = 2.2757
  Most likely break index (observation): 241

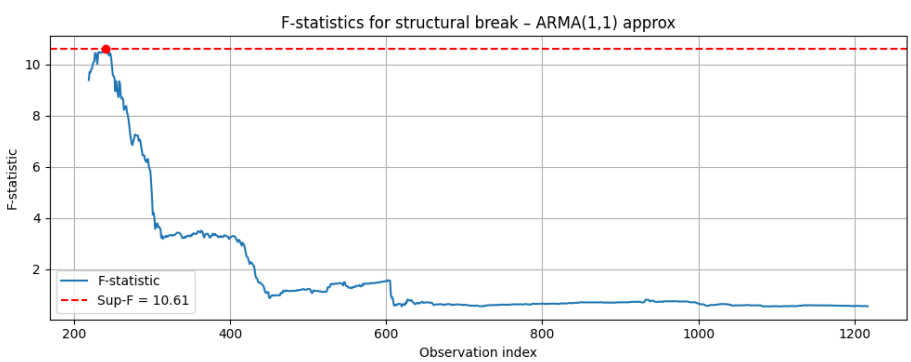

Figure 7: ARMA(1,1)

- **Testing ARMA(2,1) approx**
  → Best AR order for approximation: AR(6)
  Sup-F = 10.6055 (p-value: 0.1568)
  Ave-F = 1.7339

Exp-F = 2.2757
Most likely break index (observation): 241

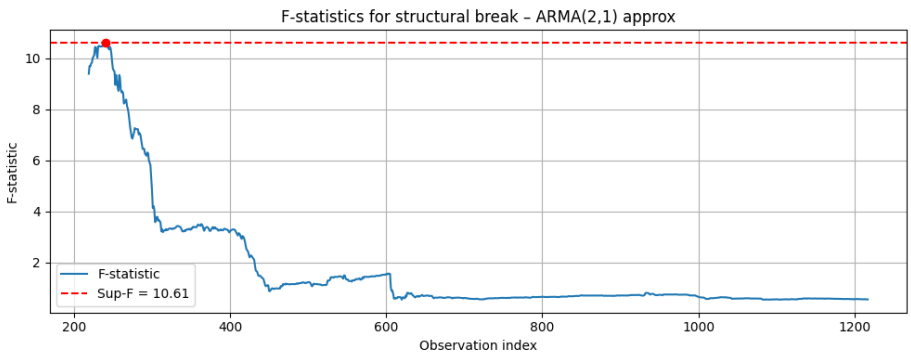

Figure 8: ARMA(2,2)

Summary of Structural Break Tests

Model	**Sup-F (p-value)**	**Ave-F**	**Exp-F**	**Break index**
AR(1)	33.0380 (0.0000)	6.7089	12.7966	259
AR(4)	14.2366 (0.0142)	2.3514	3.8083	241
AR(5)	12.3869 (0.0539)	2.0159	2.9936	241
ARMA(1,1) approx	10.6055 (0.1568)	1.7339	2.2757	241
ARMA(2,1) approx	10.6055 (0.1568)	1.7339	2.2757	241

Table 1: Summary of structural break tests. For ARMA approximations, the best AR order was AR(6).

Task 3

Experiment A: Only intercept changes

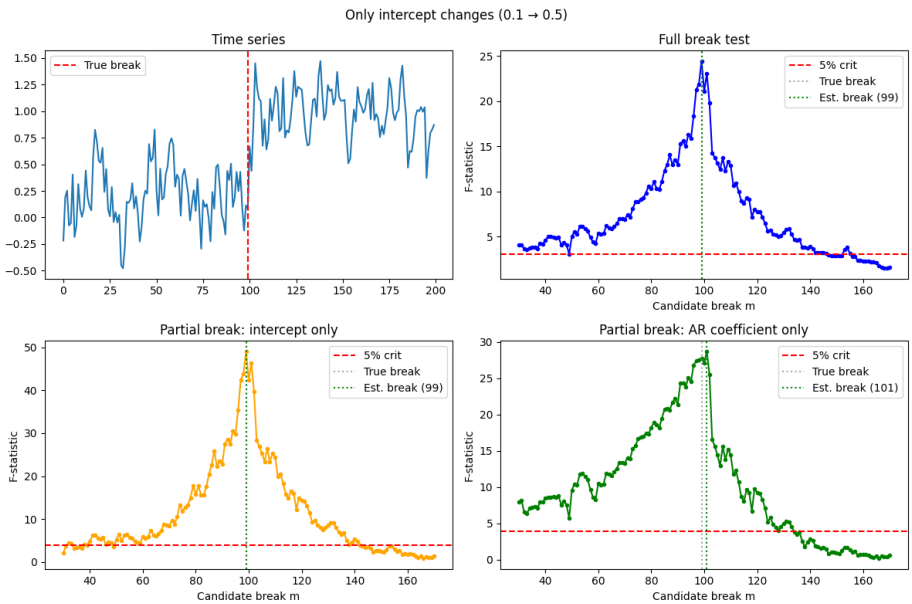

Figure 9: Only intercept changes

Experiment B: Only AR coefficient changes

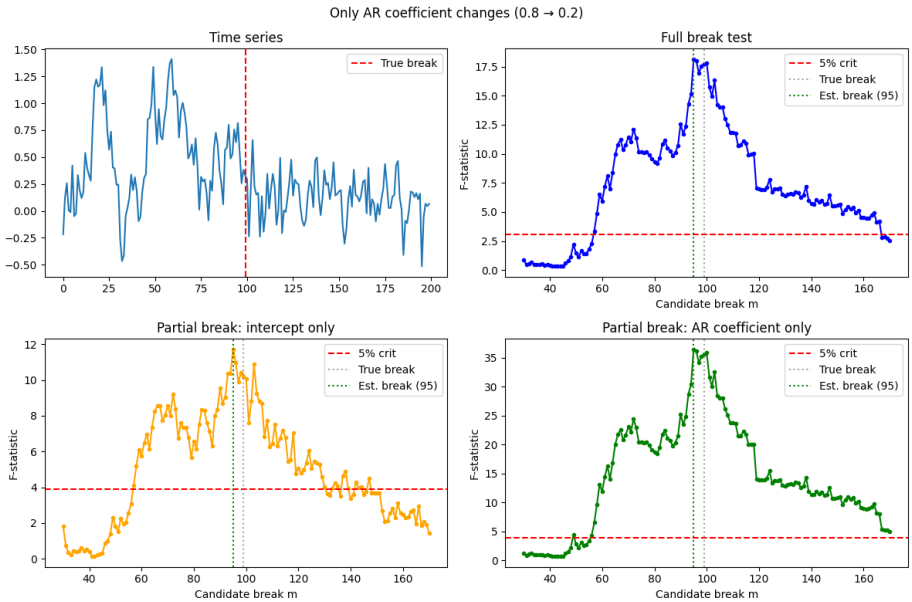

Figure 10: Only AR coefficient changes

Experiment C: Small intercept change $(0.1 \to 0.15)$

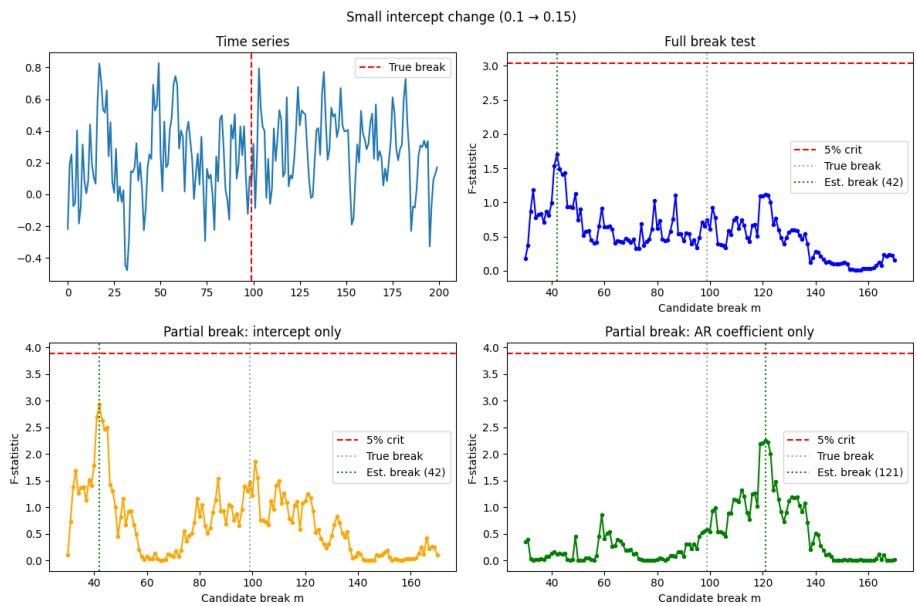

Figure 11: Small intercept change

Experiment D: Longer series ($T = 400$, AR change)

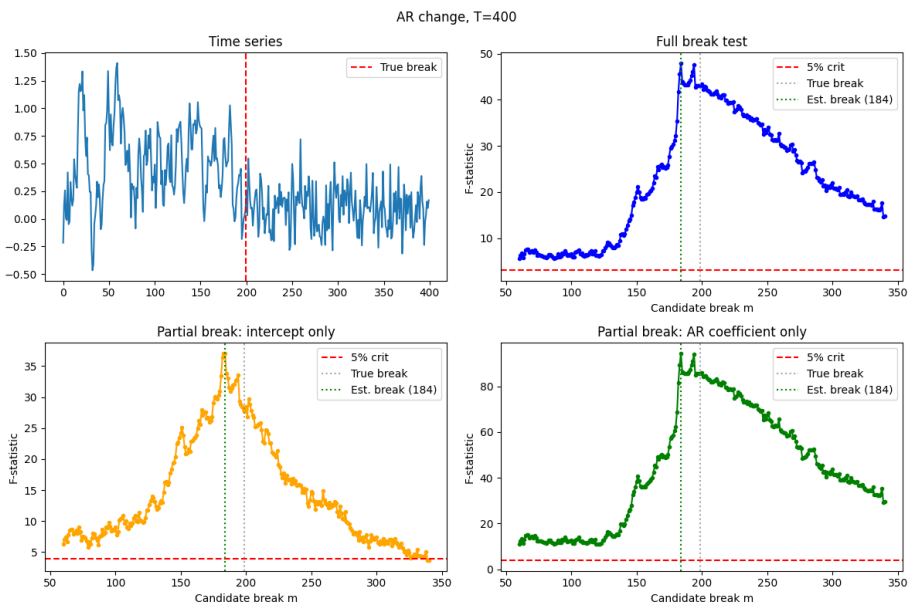

Figure 12: Longer series

Task 4

This section reports the estimated break points for different regime configurations, noise levels ($\sigma = 0.1$ and $\sigma = 0.5$), and series lengths. For each configuration, two independent replications (Rep 1 and Rep 2) are performed. The true break points are at positions 99 and 199 (i.e., after 100 observations in each regime) unless noted otherwise.

Experiments with $\sigma = 0.1$

Regime 1: $(\phi = 0.8,\ c = 0.1,\ n = 100) \to (\phi = 0.2,\ c = 0.1,\ n = 100) \to (\phi = 0.6,\ c = 0.1,\ n = 100)$

True breaks at: $[99,\ 199]$

- **Total length: 200**
  Rep 1: estimated breaks $= [7, 15, 21, 45, 66, 119]$
  Rep 2: estimated breaks $= [66, 134]$

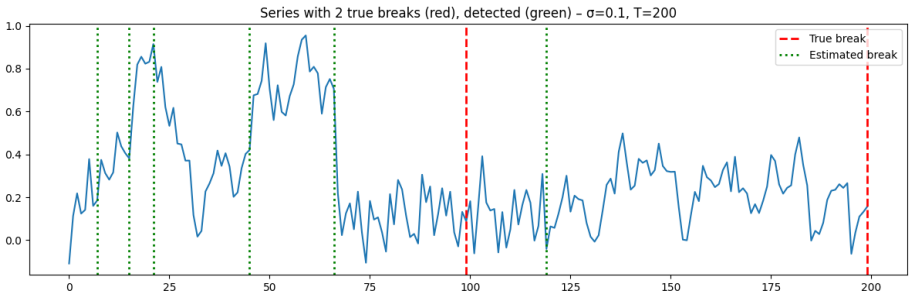

Figure 13: Enter Caption

- **Total length: 400**
  Rep 1: estimated breaks = $[7, 15, 21, 45, 66, 129, 290, 315]$
  Rep 2: estimated breaks = $[66, 132, 286]$

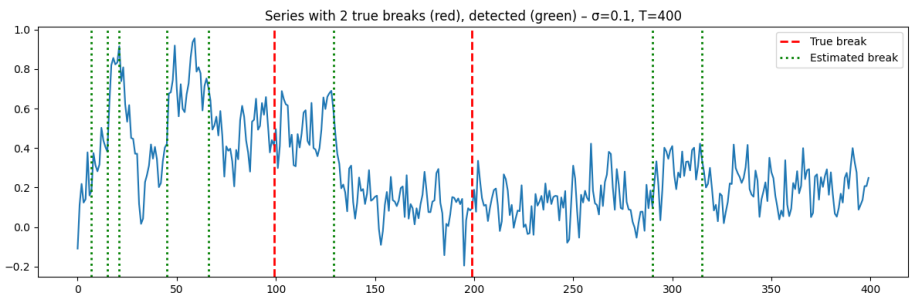

Figure 14: Enter Caption

- **Total length: 800**
  Rep 1: estimated breaks = $[266, 438, 533, 760, 769]$
  Rep 2: estimated breaks = $[66, 245, 253, 269, 292, 385, 499, 550]$

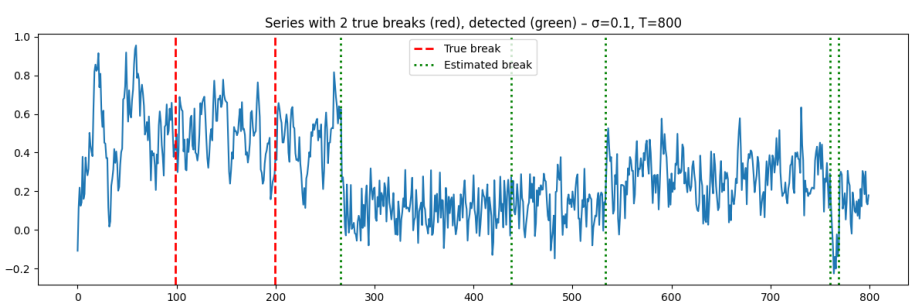

Figure 15: Enter Caption

Regime 2: $(\phi = 0.5,\ c = 0.1,\ n = 100) \to (\phi = 0.5,\ c = 0.15,\ n = 100) \to (\phi = 0.5,\ c = 0.2,\ n = 100)$

True breaks at: $[99,\ 199]$

- **Total length: 200**
  Rep 1: estimated breaks $= [16, 42, 121]$
  Rep 2: estimated breaks $= [66, 131]$

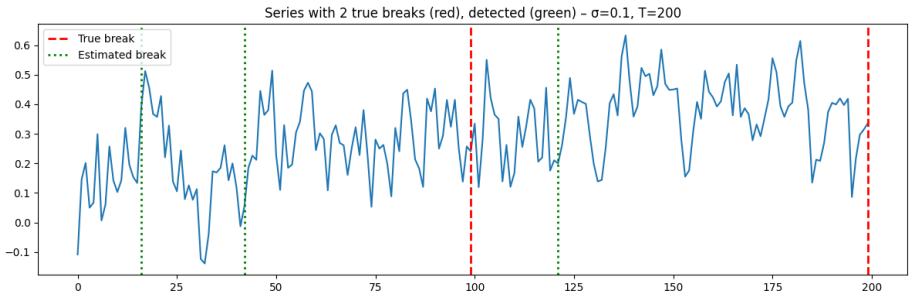

Figure 16: Enter Caption

- **Total length: 400**
  Rep 1: estimated breaks $= [132, 151, 248]$
  Rep 2: estimated breaks $= [131, 286]$

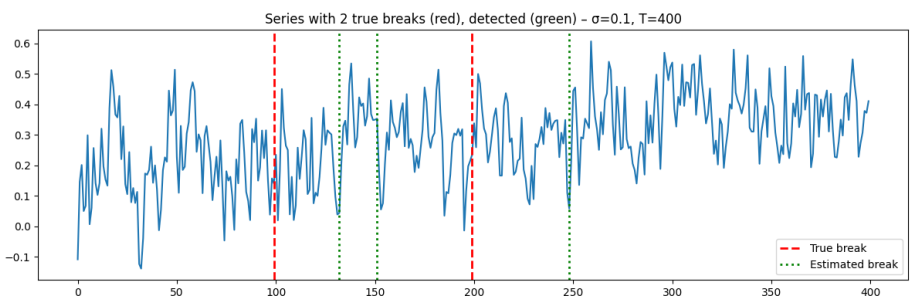

Figure 17: Enter Caption

- **Total length: 800**
  Rep 1: estimated breaks $= [286, 295, 315, 430, 448, 531, 757, 761, 769]$
  Rep 2: estimated breaks $= [66, 286, 386, 546]$

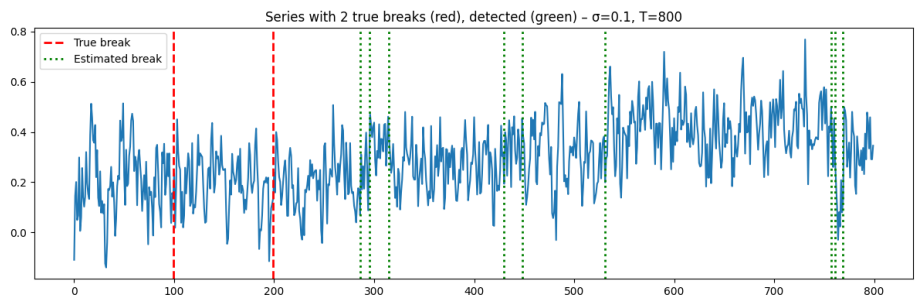

Figure 18: Enter Caption

Task 5

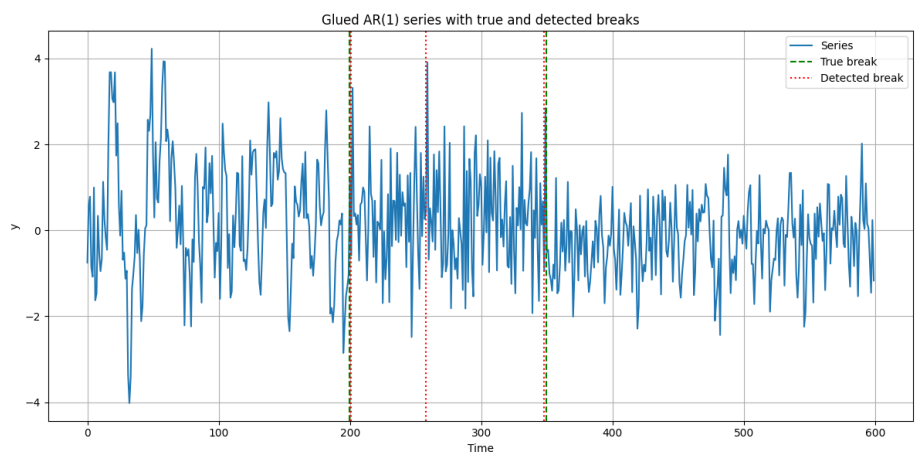

Figure 19: Enter Caption

Task 6: Nikkei

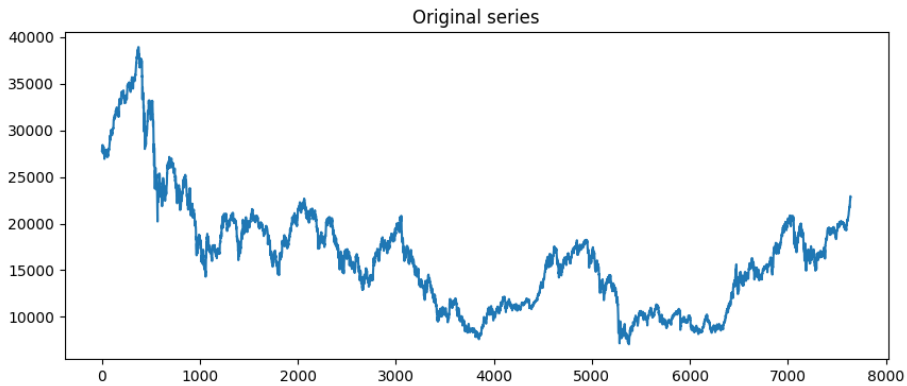

Figure 20: Enter Caption

Table 2: Stationarity Test Results (significance level $\alpha = 0.05$)

Series	**Test**	**Statistic**	**p-value**	**Lags**	**Conclusion**
4*Original series	ADF	-2.0367	0.2707	35	Non-stationary
	KPSS	6.1778	0.0100	–	Non-stationary
	PP	-2.0042	0.2848	36	Non-stationary
	Overall: Likely NON-STATIONARY (0/3 tests reject non-stationarity)				
4*Log series	ADF	-1.9111	0.3269	2	Non-stationary
	KPSS	5.8619	0.0100	–	Non-stationary
	PP	-1.8991	0.3325	36	Non-stationary
	Overall: Likely NON-STATIONARY (0/3 tests)				
4*First differences	ADF	-14.4093	0.0000	34	Stationary
	KPSS	0.2778	0.1000	–	Stationary
	PP	-88.3038	0.0000	36	Stationary
	Overall: Likely STATIONARY (3/3 tests)				
4*Log differences	ADF	-65.0002	0.0000	1	Stationary
	KPSS	0.2467	0.1000	–	Stationary
	PP	-89.7592	0.0000	36	Stationary
	Overall: Likely STATIONARY (3/3 tests)				

Note: Critical values for the ADF test (same for all series):
1%: -3.4312, 5%: -2.8619, 10%: -2.5670.
KPSS test: p-value ≤ 0.01 indicates rejection of stationarity at 5% level.
PP test uses the same critical values as ADF.

SARIMAX Results

Dep. Variable:	Nikkei
Model:	ARIMA(2, 0, 0)
Date:	Mon, 02 Mar 2026
Time:	01:23:16
Sample:	0 – 7640
No. Observations:	7640
Log Likelihood:	–52388.086
AIC:	104784.173
BIC:	104811.937
HQIC:	104793.698
Covariance Type:	opg

Variable	coef	std err	z	P> |z|	[0.025	0.975]
const	-0.6449	2.595	-0.249	0.804	-5.731	4.441
ar.L1	-0.0108	0.007	-1.496	0.135	-0.025	0.003
ar.L2	-0.0424	0.007	-5.816	0.000	-0.057	-0.028
sigma2	5.29e+04	404.000	130.930	0.000	5.21e+04	5.37e+04

Diagnostic Tests

Ljung-Box (L1) (Q):	0.00	Prob(Q):	1.00
Jarque-Bera (JB):	17190.62	Prob(JB):	0.00
Heteroskedasticity (H):	0.45	Prob(H) (two-sided):	0.00
Skew:	-0.10	Kurtosis:	10.35

SARIMAX(2,0,0) estimation results for the Nikkei series

Note: Covariance matrix calculated using the outer product of gradients (complex-step).

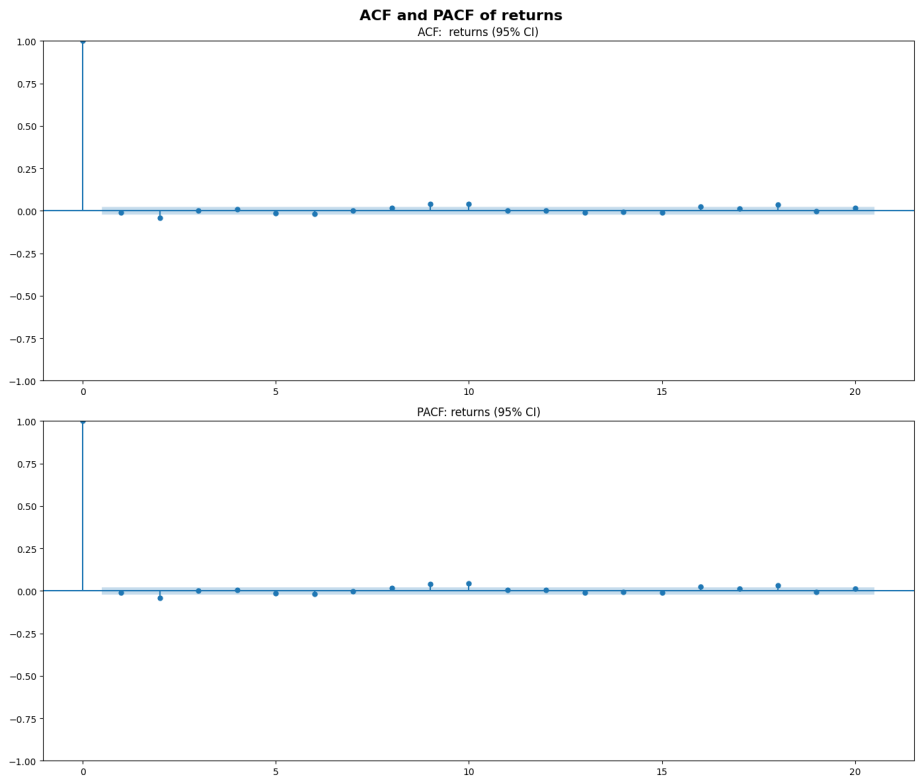

Figure 21: Enter Caption

=== Sup-F, Ave-F, Exp-F tests (AR(2) model) === Sup-F = 13.5001 Ave-F = 5.7263 Exp-F = 3.7715 Most likely break at observation index: 1151 Corresponding date: 1992-12-28 00:00:00

=== Partial break test (break only in constant) === F-statistic = 0.0197, p-value = 0.9805

=== Multiple breaks (sequential Sup-F) === Detected breaks at indices (in original returns): [np.int64(1150)] Corresponding dates: [Timestamp('1992-12-29 00:00:00')]

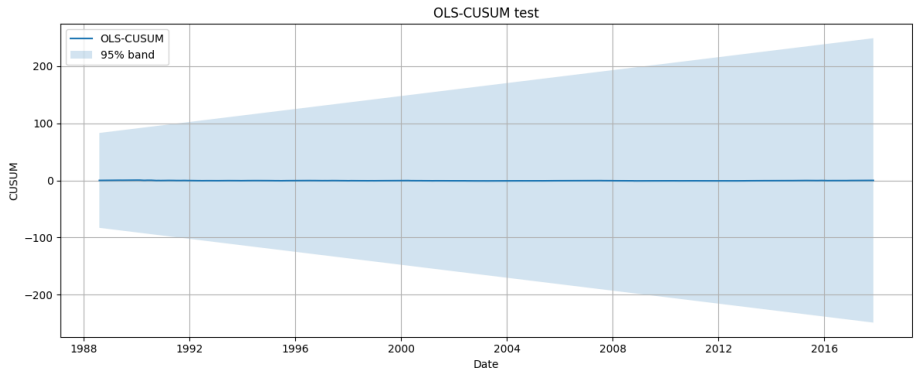

Figure 22: Enter Caption

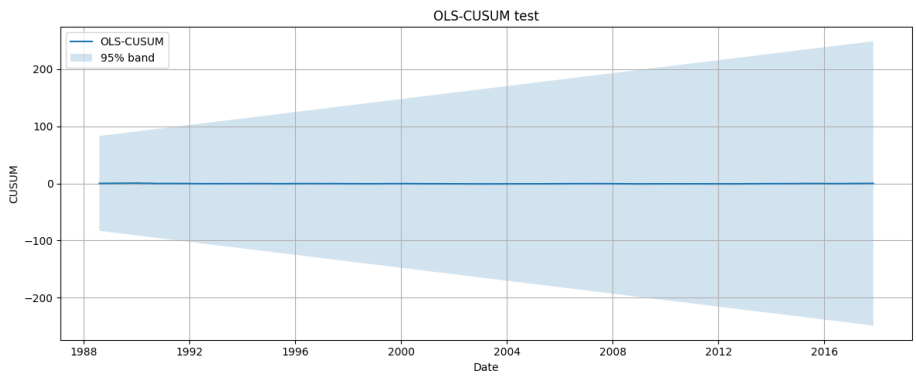

Figure 23: Enter Caption

Figure 24: Enter Caption

=== GARCH adjustment === ARCH LM test: p-value = 0.0000

Constant Mean - GARCH Model Results ==========
Dep. Variable: y R-squared: 0.000 Mean Model: Constant Mean Adj. R-squared: 0.000 Vol Model: GARCH Log-Likelihood: -51172.6 Distribution: Normal AIC: 102353. Method: Maximum Likelihood BIC: 102381. No. Observations: 7640 Date: Mon, Mar 02 2026 Df Residuals: 7639 Time: 01:51:29 Df Model: 1 Mean Model ==========
coef std err t P¿—t— 95.0———————————————————————— mu 6.2094 2.008 3.093 1.982e-03 [2.275, 10.144] Volatility Model ==========
coef std err t P¿—t— 95.0———————————————————————- omega 346.4330 106.058 3.266 1.089e-03 [1.386e+02,5.543e+02] alpha[1] 0.0820 9.967e-03 8.228 1.911e-16 [6.247e-02, 0.102] beta[1] 0.9151 9.889e-03 92.540 0.000 [0.896, 0.934] ==========

Covariance estimator: robust

— Sup-F, Ave-F, Exp-F on volatility-adjusted series — Sup-F = 2.0460, Ave-F = 0.9282, Exp-F = 0.4889

=== Forecasting next 30 days ===

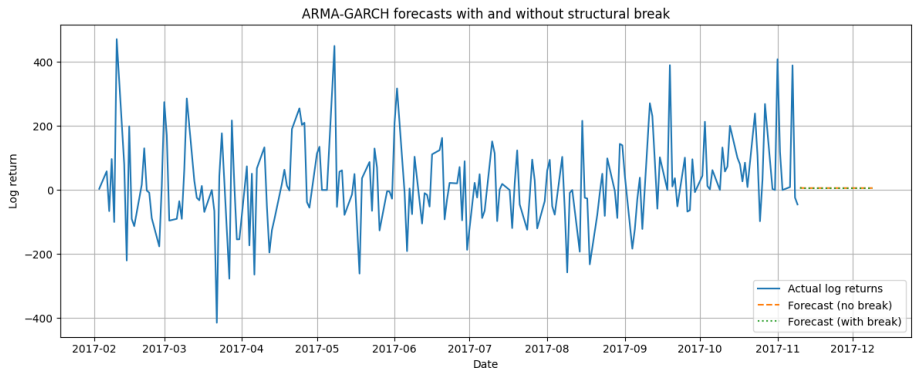

Figure 25: Enter Caption